

# YTU Mathematical Engineering

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# İçindekiler



# Bölüm 1

## Advanced Calculus - 1

### 1.1

### 1.2 Hafta 1: Vektörler ve Temel Kavramlar

**Definiton 1.1.** A function of two variables is a rule that assigns a real number  $W$  sometimes write:  $f : D \subset \mathbb{R}^2 \rightarrow \mathbb{R}$ ; for instance,  $f(x, y) = x^2 + y^2$  or  $g(x, y) = x^2 - e^y$ .

Likewise, a function of three variables is a rule that assigns a real number to each ordered triple  $(x, y, z)$  in its domain  $D \subset \mathbb{R}^3$ . We sometimes write:  $f : D \subset \mathbb{R}^3 \rightarrow \mathbb{R}$ ; for instance,  $f(x, y, z) = x^2 + y^2 + z^2$  or  $g(x, y, z) = x^2 - e^{y+z}$ .

$f : D \subset \mathbb{R}^n \rightarrow \mathbb{R}$  is a function of  $n$  variables if it assigns a real number to each ordered  $n$ -tuple  $(x_1, x_2, \dots, x_n)$  in its domain  $D \subset \mathbb{R}^n$ .

**Example.** (Lesson: Advanced Calculus - 1) (Week: 1) (Topic: )

Find the domain of the functions.

Example 1.1  $f(x, y) = \ln 4 - x^2 - 4y^2$

Example 1.2  $g(x, y) = \sqrt{x^2 + y^2 - 1} + \sqrt{4 - x^2 - y^2}$

Example 1.3  $f(x, y) = \frac{1}{\sqrt{4x^2 - y^2}}$

**Definiton 1.2.** A neighborhood of a point  $\mathbf{a} \in \mathbb{R}^n$  is a set that contains all points  $\mathbf{x}$  that are "close" to  $\mathbf{a}$ . More formally, we can define a neighborhood of  $\mathbf{a}$  as follows:

Let  $r > 0$ . The open ball of radius  $r$  centered at  $\mathbf{a}$  is defined as:

$$B_r(\mathbf{a}) = \{\mathbf{x} \in \mathbb{R}^n : \|\mathbf{x} - \mathbf{a}\| < r\}$$

A neighborhood of  $\mathbf{a}$  is any set that contains an open ball centered at  $\mathbf{a}$ .

**Definiton 1.3.** The interior of a set  $A \subset \mathbb{R}^n$ , denoted  $\text{int}(A)$ , is the largest open set contained in  $A$ . In other words, a point  $\mathbf{x} \in A$  is in the interior of  $A$  if there exists a neighborhood of  $\mathbf{x}$  that is entirely contained in  $A$ .

**Definiton 1.4.** The boundary of a set  $A \subset \mathbb{R}^n$ , denoted  $\partial A$ , is the set of points that can be approached both from inside  $A$  and from outside  $A$ . Formally, a point  $\mathbf{x} \in \mathbb{R}^n$  is in the boundary of  $A$  if every neighborhood of  $\mathbf{x}$  contains at least one point in  $A$  and at least one point not in  $A$ .